Pembroke Hill Invitational - Repaired practice edition

Division B | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

35. (1 points) Authored replacement for source item 35: For a simple solenoid with fixed current and fixed core geometry, does increasing the number of turns per unit length increase or decrease its magnetic field? Explain.

Answer: _____
_____
_____

repair-case. (2 points) Authored written substitute for unavailable source material: Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. No

B. Always

C. Only for any red LED

D. Only at 5 V

Answer: _____